

Alfred Patisenah

Scientist | IBO Coach

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Summary

Molecular Microbiologist with expertise in clinical diagnostics, CRISPR-based gene editing, and bioinformatics. Proven track record of developing automated analysis pipelines and managing international research collaborations. Passionate about science education. Collaborates with Indonesian government in coaching Indonesian IBO delegates and training the next generation of Indonesian biologist.

Experience

Prodia Clinical Laboratory , Advanced Microbiology Scientist - Evaluated medical devices and methods for culture-based identification, antimicrobial susceptibility testing, and 17 NAATs. - Managed the training program for lab technologists on newly launched tests. - Collaborated with international partner to bring gut microbiome testing to Indonesia. - Developed a Python pipeline for automated qPCR analysis, improving interpretation efficiency.	Jakarta, Indonesia July 2024 – present 2 years 4 months
Tim Olimpiade Biologi Indonesia , Scientific Committee Member - Collaborated with experts and the ministry of education to hold national biology olympiad with over 20,000 participants. - Spearheaded the formation of Indonesian biology olympiad alumni community to educate impoverished students.	Bandung, Indonesia 2018 – present 8 years
Institut Teknologi Bandung , Research Assistant - Worked on <i>Rhodotorula</i> yeast metabolic engineering to enhance PUFA production. - Performed plasmid construction and optimized <i>Agrobacterium</i>-mediated transformation.	Bandung, Indonesia Feb 2024 – June 2024 5 months
Locavore NXT , Fermentation RnD Implemented submerged fermentation method to enhance vinegar production.	Bali, Indonesia Nov 2023 – Feb 2024 4 months
Institut Teknologi Del , Intern Researcher Optimized substrate formulation for <i>A. niger</i> solid-state fermentation to enhance lipase production.	Balige, Indonesia June 2021 – Aug 2021 3 months

Education

MSc Institut Teknologi Bandung , Biotechnology - Investigated the efficacy of bicistronic chaperone-substrate co-localized expression system to enhance recombinant Cas9 solubility. - Secured Rp150m research funding.	Bandung, Indonesia 2022 – 2023
BSc Institut Teknologi Bandung , Microbiology - Produced CRISPR-Cas9 ribonucleoprotein for transgene-free chili genome editing. - Teaching assistant for Research Design, Computation Lab, and Molecular Biology Lab courses.	Bandung, Indonesia 2018 – 2022

Publications

sgRNA Design and in vitro Nucleolytic Analysis of Cas9-RNP Complex for Transgene-Free Genome Editing of eIF4E1 Gene from Capsicum annuum L.

Dec 2023

Tham, J., **Patisenah, A.**, Tjia, T.O.S., Signorelli, S., Taufik, I., Meitha, K.

journal.ugm.ac.id/ijbiotech/article/view/86778/0 (Indonesian Journal of Biotechnology)

Skills

Molecular Biology: DNA/RNA/protein isolation, cloning, PCR/RT-PCR/OE-PCR, protein purification, SDS-PAGE, BSL-2 pathogen assays

Bioinformatics and Programming: Python, ImageJ, BLAST, MEGA11, SnapGene, Benchling

Clinical Diagnostics: IVD devices (MALDI Biotyper, BD Phoenix, Vitek 2, cobas 6800, GeneXpert, HybriSpot)

Languages

Indonesian

Native speaker

English

Advanced (C1)

Hokkien

Fluent

Awards

Dean's List

School of Life Sciences and Technology, ITB

2023

Ganesha Talent Assistanship

Institut Teknologi Bandung

2022

Ganesha Karsa Award

Institut Teknologi Bandung

2022

Research Program Funds, LPPM ITB

Institut Teknologi Bandung

2022

Winner of XVI NCRM NICHE Fujio Cup

NCRM NICHE

2021

Silver Medal in Biology, 16th National Science Olympiad

Indonesian Ministry of Education

2017